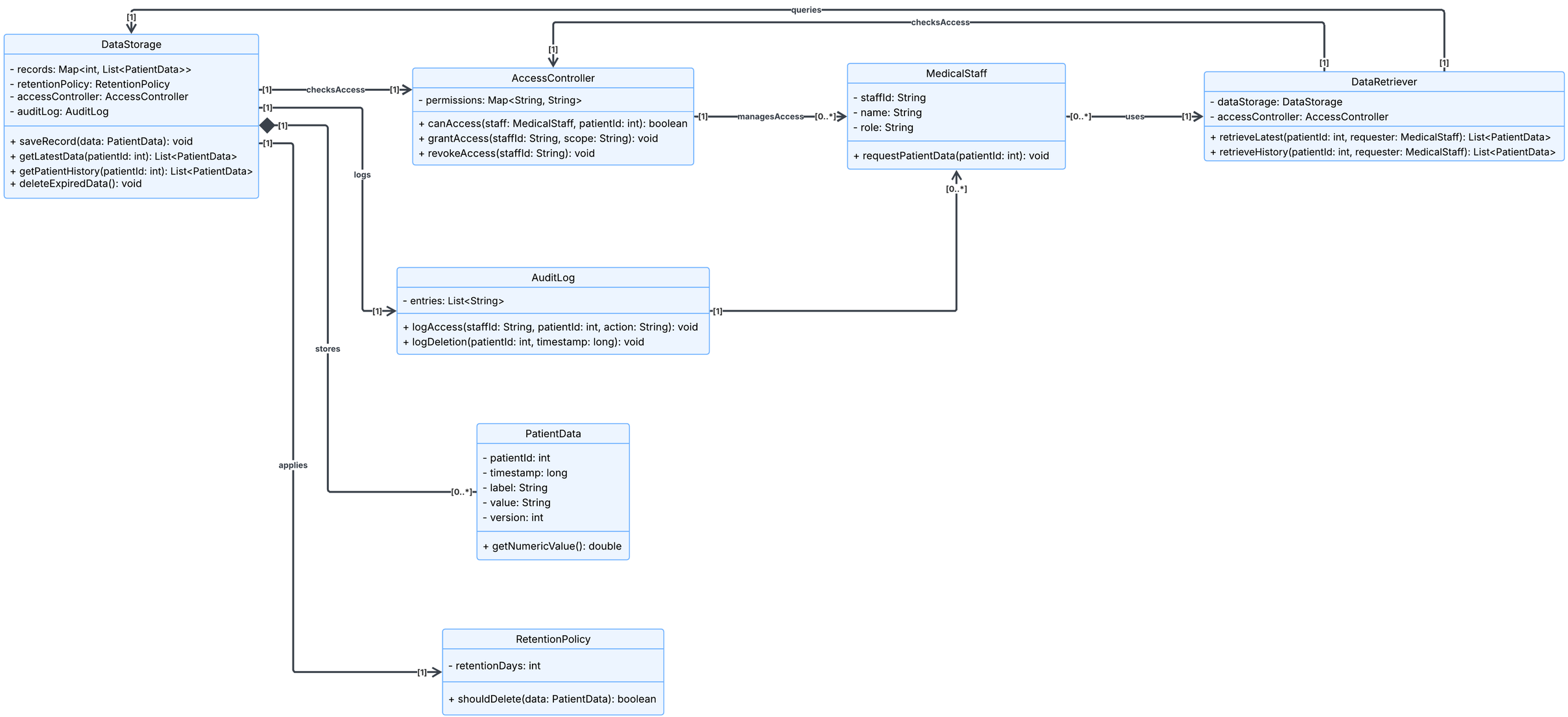
CHMS Data Storage System — Class Diagram



This diagram models how the CHMS stores, protects, and retrieves incoming patient data. The design separates storage, retrieval, access control, retention, and auditing so the subsystem remains secure, maintainable, and easy to extend.

DataStorage is the central storage component. It stores PatientData records and provides methods to retrieve the latest measurements or patient history. PatientData represents one time-stamped vital record and includes a version field to support traceability. A composition relationship is used between DataStorage and PatientData because the records are managed as part of the storage lifecycle.

DataRetriever is responsible for querying stored data on behalf of medical staff. This avoids letting users access the storage component directly. AccessController checks whether a staff member is allowed to access a specific patient's data, which reflects privacy and security requirements. MedicalStaff is modeled separately as the requester, not as part of the storage logic itself.

RetentionPolicy defines when old records should be deleted, keeping deletion rules separate from storage operations. AuditLog records important actions such as access and deletion events, improving accountability and traceability. Overall, the subsystem emphasizes secure retrieval, controlled access, and clear separation of responsibilities, which supports a realistic hospital data management design.